

Disproof of Conjecture 00000002913

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: a degree- d polynomial map on $\mathbb{F}_q$ is injective when $q > d^2$ (finite-field injectivity threshold spectrum).

2 The refutation

Take $d = 2$, $q = 5$: the claimed threshold holds ($5 > 4$), yet the degree-2 map $x \mapsto x^2$ on $\mathbb{F}_5$ is not injective:

$$4^2 = 16 = 1 = 1^2 \pmod{5}, \quad 4 \neq 1$$

(kernel-certified collision). In fact every square in $\mathbb{F}_5$ lands in $\{0, 1, 4\}$: $0^2 = 0$, $1^2 = 1$, $2^2 = 4$, $3^2 = 9 = 4$, $4^2 = 16 = 1$ — the image has $3 < 5$ elements. A complete enumeration shows more: *no* quadratic map ($a \neq 0$) on $\mathbb{F}_5$ (all 100) or $\mathbb{F}_7$ (all 294) is injective — a non-degenerate quadratic over $\mathbb{F}_q$ (q odd) has image of size at most $(q+1)/2 < q$ — while the threshold $q > d^2 = 4$ holds for both fields. The plane version fails identically: $(x, y) \mapsto (x^2, y^2)$ on $\mathbb{F}_5^2$ has image $9 < 25$ with $(1, 0)$ and $(4, 0)$ colliding.

3 Verification

`reproduce.py` exhibits the certified collision and 3-element image, enumerates all quadratic maps on $\mathbb{F}_5$ and $\mathbb{F}_7$ (all non-injective, max image size $(q+1)/2$), and checks the plane instance. The Lean package (core Lean 4, v4.33.1) certifies `threshold_holds` ($5 > 4$), `collision` ($4 \neq 1$ and $4^2 \equiv 1^2 \pmod{5}$), `image_three` (all five inputs into the pairwise-distinct set $\{0, 1, 4\}$, $3 < 5$), and `conjecture_refuted`. All four audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the exact residue arithmetic at the first instance; the enumeration over all quadratic maps and the plane instance are exhaustive script computations. The threshold claim is refuted at $d = 2$, $q = 5$; the conjecture’s “Chebyshev/Lagrange inversion” clause is not needed for the refutation.